

Facts and Statistics in the IMO

Version 1.2

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Introduction

Everyone reading this should know what the IMO is. But how much do you know about the IMO and the Hong Kong team? As a team member or a trainee of the Hong Kong team, we should learn more about the facts in the IMO!

Most of the data used in this document are found from the IMO official website imo-official.org and the IMOHKC website www.imohkc.org.hk, the latter of which is not up-to-date. Since many of the results given in this document are obtained by analyzing the existing data manually, it is not guaranteed that all information is accurate. Nevertheless, the major purpose of this document is to show you some interesting facts and records which are rarely known.

In section 1, we first use this occasion to list out all Hong Kong team members since our first participation in the IMO. Also, we will see the performances of all schools which have students joining the IMO. As far as I know, this is the only place which contains statistics of this kind.

In section 2, we will go through some facts and history in the IMO. Apart from some basic introductions to the IMO such as the format of the contest, there is also less well-known information such as the process of coordination, and details of some solutions which received special prizes.

Lastly, in sections 3 and 4, we provide various kinds of statistics in the IMO which cannot be easily found, including the performances of different countries, the performances in different problems, and the performances of the Hong Kong team members, etc.

Enjoy, and see if you can be the next one to break some of the records here!

(Note: New records in this version are put in [purple](#).)

1 The Hong Kong Team

First, let us introduce the Hong Kong team members since the first year, 1988, we joined the IMO. All of them deserve a big clap!

Notes:

- Some information in the old days cannot be found, and some information from different sources is inconsistent.
- In this document, we shall label the team members in each year from HKG1 to HKG6 according to the order in this list. However, this does not necessarily represent the contestant number in that year. In fact, this is only the order given on the official website.

Year	Team Member	School
1988	Chan Shun	Queen's College
	Choy Shu Hung	King's College
	Leung Wing Hong	Queen Elizabeth School
	Tai Wai Ling	Diocesan Boys' School
	Yau Shuk Han	St. Paul's Co-educational College
	Yip Nung Kwan	Tsuen Wan Government Secondary School
1989	Chan Kai Pak	St. Paul's Co-educational College
	Chiu Shin Yeung	King's College
	Choy Shu Hung	King's College
	Man Lai Chee	St. Paul's Co-educational College
	Tam Ting Kin	Kwun Tong Government Secondary School
	Yau Shuk Han	St. Paul's Co-educational College
1990	Chan Kin Wah	TWGHs Mrs. Wu York Yu Memorial College
	Lau Chi Hin	Pentecostal School
	Ho Yuk	King's College
	Leung Wing Kai	Queen's College
	Cheng Wing Leung	Ha Kwai Chung Government Secondary Technical School
	Ma Jim Lok	King's College
1991	Chan Kin Wah	TWGHs Mrs. Wu York Yu Memorial College
	Cheung Trevor Cheuk Wah	Queen's College
	Ko Thomas Chi Kin	Queen's College
	Ngai Chi Ho	Diocesan Boys' School
	To Kar Keung	King's College
	Wong Wilkie	Diocesan Boys' School

Year	Team Member	School
1992	Chung Wai Yin	King's College
	Lam Chi Wai	CCC Chuen Yuen College
	Lam Pei Fung	Queen's College
	Lee Wai Fun	St. Mark's School
	Suen Yun Leung	King's College
	To Kar Keung	King's College
1993	Chu Hoi Pan	Chong Gene Hang College
	Chan Tsz Ho	Ying Wa College
	Lam Chi Wai	CCC Chuen Yuen College
	Lin Kwong Shing	Tsuen Wan Government Secondary School
	Tsui Ka Hing	Queen Elizabeth School
	Yung Fai	CCC Chuen Yuen College
1994	Chu Hoi Pan	Chong Gene Hang College
	Ho Wing Yip	Clementi Secondary School
	Poon Wai Hoi Bobby	St. Paul's College
	Suen Yun Leung	King's College
	Tsui Ka Hing	Queen Elizabeth School
	Wong Him Ting	Salesian English School
1995	Ho Wing Yip	Clementi Secondary School
	Cheung Kwok Koon	SKH Bishop Mok Sau Tseng Secondary School
	Yu Chun Ling	Ying Wa College
	Mok Tze Tao Edmond	Queen's College
	Poon Wai Hoi Bobby	St. Paul's College
	Wong Him Ting	Salesian English School
1996	Ho Wing Yip	Clementi Secondary School
	Yu Chun Ling	Ying Wa College
	Law Siu Lung David	Diocesan Boys' School
	Mok Tze Tao Edmond	Queen's College
	Poon Wai Hoi Bobby	St. Paul's College
	Tse Shan Shan	Tuen Mun Government Secondary School
1997	Chan Chung Lam	Bishop Hall Jubilee School
	Cheung Pok Man	STFA Leung Kau Kui College
	Lau Lap Ming Alvin	St. Paul's College
	Leung Wing Chung	Queen Elizabeth School
	Mok Tze Tao Edmond	Queen's College
	Yu Ka Chun	Queen's College
1998	Chan Kin Hang	Bishop Hall Jubilee School
	Cheung Pok Man	STFA Leung Kau Kui College
	Choi Ming Cheung	King's College
	Lau Lap Ming Alvin	St. Paul's College
	Law Ka Ho	Queen Elizabeth School
	Leung Wing Chung	Queen Elizabeth School

Year	Team Member	School	Form
1999	Chan Ho Leung	Diocesan Boys' School	
	Chan Kin Hang	Bishop Hall Jubilee School	
	Chan Tsz Hong Hubert	Diocesan Boys' School	
	Law Ka Ho	Queen Elizabeth School	
	Ng Ka Wing	STFA Leung Kau Kui College	
	Wong Chun Wai Carto	Choi Hung Estate Catholic Secondary School	
2000	Chan Kin Hang	Bishop Hall Jubilee School	S6
	Fan Wai Tong	St. Mark's School	S7
	Law Ka Ho	Queen Elizabeth School	S7
	Ng Ka Wing	STFA Leung Kau Kui College	S7
	Wong Chun Wai Carto	Choi Hung Estate Catholic Secondary School	S7
	Yu Hok Pun	SKH Bishop Baker Secondary School	S4
2001	Chan Kin Hang	Bishop Hall Jubilee School	S7
	Chao Khek Lun Harold	St. Paul's College	S6
	Cheng Kei Tsi Daniel	La Salle College	S6
	Ko Man Ho	Wah Yan College, Kowloon	S7
	Leung Wai Ying	Queen Elizabeth School	S6
	Yu Hok Pun	SKH Bishop Baker Secondary School	S5
2002	Chao Khek Lun Harold	St. Paul's College	S7
	Chau Suk Ling	Queen Elizabeth School	S7
	Cheng Kei Tsi Daniel	La Salle College	S7
	Ip Chi Ho Ivan	St. Joseph's College	S6
	Leung Wai Ying	Queen Elizabeth School	S7
	Yu Hok Pun	SKH Bishop Baker Secondary School	S6
2003	Chung Tat Chi	Queen Elizabeth School	S6
	Kwok Tsz Chiu	Yuen Long Merchants Association Secondary School	S5
	Lau Wai Shun	Tsuen Wan Public Ho Chuen Yiu Memorial College	S7
	Siu Tsz Hang	STFA Leung Kau Kui College	S7
	Yeung Kai Sing	La Salle College	S6
	Yu Hok Pun	SKH Bishop Baker Secondary School	S7

Year	Team Member	School	Form
2004	Cheung Yun Kuen	Hong Kong Chinese Women's Club College	S6
	Chung Tat Chi	Queen Elizabeth School	S7
	Kwok Tsz Chiu	Yuen Long Merchants Association Secondary School	S6
	Poon Ming Fung	STFA Leung Kau Kui College	S6
	Tang Chiu Fai	HKTA Tang Hin Memorial Secondary School	S6
	Wong Hon Yin	Queen's College	S5
2005	Chung On Yip	Queen Elizabeth School	S4
	Siu Ho Chung	Queen's College	S6
	Hui Jack	Queen's College	S5
	Poon Ming Fung	STFA Leung Kau Kui College	S7
	Tsoi Yun Pui	Sha Tin Government Secondary School	S6
	Wong Chiu Wai	St. Joseph's College	S3
2006	Chung On Yip	Queen Elizabeth School	S5
	Fong Pok Wai	SKH Lam Woo Memorial Secondary School	S5
	Hui Jack	Queen's College	S6
	Tsoi Yun Pui	Sha Tin Government Secondary School	S7
	Wong Chiu Wai	St. Joseph's College	S4
	Yuen Chak Hei	Queen's College	S6
2007	Fong Pok Wai	SKH Lam Woo Memorial Secondary School	S6
	Hui Jack	Queen's College	S7
	Lam Cho Ho	STFA Lee Shau Kee College	S5
	Lee Pak Hin	Queen's College	S5
	Li Cheuk Ting	La Salle College	S5
	Wong Chiu Wai	St. Joseph's College	S5
2008	Chung Ping Ngai	La Salle College	S4
	Lam Cho Ho	STFA Lee Shau Kee College	S6
	Li Cheuk Ting	La Salle College	S6
	Tai Wai Ming	SKH Leung Kwai Yee Secondary School	S7
	Tai Yee Ka	St. Paul's College	S7
	Wong Chiu Wai	St. Joseph's College	S6
2009	Ching Tak Wing	Queen's College	S5
	Chung Ping Ngai	La Salle College	S5
	Hung Ka Kin Kenneth	Diocesan Boys' School	S5
	Lam Cho Ho	STFA Lee Shau Kee College	S7
	Ng Ngai Fung	STFA Leung Kau Kui College	S6
	Yu Tak Hei	La Salle College	S4

Year	Team Member	School	Form
2010	Ching Tak Wing	Queen's College	S6
	Chung Ping Ngai	La Salle College	S6
	Hung Ka Kin Kenneth	Diocesan Boys' School	S6
	Tam Ka Yu	Queen's College	S5
	Yip Hok Pan	Ying Wa College	S7
	Yu Tak Hei	La Salle College	S5
2011	Ching Tak Wing	Queen's College	S7
	Chow Chi Hong	Bishop Hall Jubilee School	S6
	Li Yau Wing	Ying Wa College	S5
	Lo Chun Tung Raymond	King's College	S6
	Tam Ka Yu	Queen's College	S6
	Wo Bar Wai Barry	La Salle College	S6
2012	Chow Chi Hong	Bishop Hall Jubilee School	S7
	Kung Man Kit	SKH Lam Woo Memorial Secondary School	S6
	Lau Chun Ting	St. Paul's Co-educational College	S3
	Li Yau Wing	Ying Wa College	S6
	Loo Andy	St. Paul's Co-educational College	S6
	Wong Sze Nga	Diocesan Girls' School	S4
2013	Hui Pak Nam	La Salle College	S5
	Lau Chun Ting	St. Paul's Co-educational College	S4
	Li Jianhui	CNEC Christian College	S6
	Man Siu Hang	CCC Ming Yin College	S6
	Wong Hing Shing	La Salle College	S5
	Wong Sze Nga	Diocesan Girls' School	S5
2014	Cheung Wai Lam	Queen Elizabeth School	S4
	Hui Pak Nam	La Salle College	S6
	Lau Chun Ting	Diocesan Boys' School	S5
	Wong Hing Shing	La Salle College	S6
	Wong Sze Nga	Diocesan Girls' School	S6
	Yu Hoi Wai	La Salle College	S3
2015	Cheung Wai Lam	Queen Elizabeth School	S5
	Kwok Man Yi	Baptist Lui Ming Choi Secondary School	S4
	Lee Shun Ming Samuel	CNEC Christian College	S4
	Tung Kam Chuen	La Salle College	S6
	Wu John Michael	Hong Kong International School	S4
	Yu Hoi Wai	La Salle College	S4

Year	Team Member	School	Form
2016	Cheung Wai Lam	Queen Elizabeth School	S6
	Kwok Man Yi	Baptist Lui Ming Choi Secondary School	S5
	Lee Shun Ming Samuel	CNEC Christian College	S5
	Leung Yui Hin Arvin	Diocesan Boys' School	S5
	Wu John Michael	Hong Kong International School	S5
	Yu Hoi Wai	La Salle College	S5
2017	Kwok Man Yi	Baptist Lui Ming Choi Secondary School	S6
	Lee Shun Ming Samuel	CNEC Christian College	S6
	Leung Yui Hin Arvin	Diocesan Boys' School	S6
	Tse Alvin Cheuk Hin	Yew Chung International School	G11
	Ye Jeff York	G. T. (Ellen Yeung) College	S6
	Yu Hoi Wai	La Salle College	S6
2018	Cheng Yan Yau	Discovery College	G13
	Fong Tsz Lo	SKH Lam Woo Memorial Secondary School	S6
	Ho Tsz Chun	HKBU Affiliated School Wong Kam Fai Secondary and Primary School	S5
	Ngai Chi Ki	Pui Ching Middle School	S6
	Tse Alvin Cheuk Hin	Yew Chung International School	G12
	Wong Tsz Chuen	La Salle College	S6
2019	Ho Sui Kei	HKTA Tang Hin Memorial Secondary School	S6
	Lee Wan	La Salle College	S5
	Leung Harris	Diocesan Boys' School	S4
	Sheremeta Daniel Weili	Hong Kong International School	S5
	Tong Nok To Omega	Pui Ching Middle School	S6
	Xu Bruce Changlong	West Island School	G13
2020	Chu Cheuk Hei	St. Paul's Co-educational College	S3
	Chui Tsz Fung	Ma Tau Chung Government Primary School	P6
	Lai Wai Lok	St. Paul's Co-educational College	S2
	Leung Harris	Diocesan Boys' School	S5
	Sheremeta Daniel Weili	Hong Kong International School	S6
	Yau Ching Yan Timothy	Ying Wa College	S4

Year	Team Member	School	Form
2021	Chu Cheuk Hei	St. Paul's Co-educational College	S4
	Chui Tsz Fung	Diocesan Boys' School	S1
	Lai Wai Lok	St. Paul's Co-educational College	S3
	Leung Harris	Diocesan Boys' School	S6
	Yau Ching Yan Timothy	Ying Wa College	S5
	Yiu Chun Hei	SKH Lam Woo Memorial Secondary School	S3
2022	Cheng Nick Hang	Diocesan Boys' School	S6
	Chu Cheuk Hei	St. Paul's Co-educational College	S5
	Hsieh Chong Ho	Tsuen Wan Government Secondary School	S5
	Lai Wai Lok	St. Paul's Co-educational College	S4
	Yau Ching Yan Timothy	Ying Wa College	S6
	Yiu Chun Hei	SKH Lam Woo Memorial Secondary School	S4
2023	Chase Sebastian	The Harbour School	G8
	Chu Cheuk Hei	St. Paul's Co-educational College	S6
	Hsieh Chong Ho	Tsuen Wan Government Secondary School	S6
	Kwan Yung Ho	Diocesan Boys' School	S3
	Ng Hok Lai	St. Paul's Co-educational College	S5
	Yiu Chun Hei	SKH Lam Woo Memorial Secondary School	S5

In these 36 years, there have been 129 different team members from 47 different schools, including 6 female team members. The following table lists all the schools which have at least one team member, together with the awards obtained.

Notes:

- ## denotes the number of team members counted with multiplicity, while # denotes the number of distinct team members.
- G, S, B and HM denote the numbers of gold medals, silver medals, bronze medals, and honourable mentions respectively.

School	##	#	G	S	B	HM
La Salle College	22	12	1	7	12	1
Queen's College	21	14	3	6	6	3
Queen Elizabeth School	18	9	0	6	8	1
Diocesan Boys' School	17	13	1	5	6 ↑	3
St. Paul's Co-educational College	15	8	2 ↑	3	6 ↑	2
King's College	12	9	0	3	6	1

School	##	#	G	S	B	HM
Ying Wa College	9	5	1	4	4	0
STFA Leung Kau Kui College	8	5	0	2	4	1
St. Paul's College	8	4	0	1	6	1
SKH Lam Woo Memorial Secondary School ↑	7	4	0	4 ↑	0	2
Bishop Hall Jubilee School	7	3	0	1	4	1
St. Joseph's College	5	2	0	2	2	0
Tsuen Wan Government Secondary School ↑	4	3	0	1	3 ↑	0
CNEC Christian College	4	2	2	0	2	0
Hong Kong International School	4	2	1	1	2	0
SKH Bishop Secondary School	4	1	1	3	0	0
CCC Chuen Yuen College	3	2	0	0	1	2
STFA Lee Shau Kee College	3	1	0	3	0	0
Clementi Secondary School	3	1	0	2	1	0
Baptist Lui Ming Choi Secondary School	3	1	0	1	2	0
Diocesan Girls' School	3	1	0	1	2	0
Pui Ching Middle School	2	2	0	0	1	1
HKTA Tang Hin Memorial Secondary School	2	2	0	0	1	0
St. Mark's School	2	2	0	0	0	1
Sha Tin Government Secondary School	2	1	2	0	0	0
Yuen Long Merchants Association Secondary School	2	1	0	2	0	0
Salesian English School	2	1	0	1	1	0
TWGHs Mrs. Wu York Yu Memorial College	2	1	0	0	2	0
Yew Chung International School	2	1	0	0	2	0
Chong Gene Hang College	2	1	0	0	1	0
Choi Hung Estate Catholic Secondary School	2	1	0	0	0	2
Kwun Tong Government Secondary School	1	1	0	1	0	0
Ma Tau Chung Government Primary School	1	1	0	1	0	0
CCC Ming Yin College	1	1	0	0	1	0
G. T. (Ellen Yeung) College	1	1	0	0	1	0
Hong Kong Chinese Women's Club College	1	1	0	0	1	0
Pentecostal School	1	1	0	0	1	0
SKH Bishop Mok Sau Tseng Secondary School	1	1	0	0	1	0
The Harbour School	1	1	0	0	1 ↑	0
Wah Yan College, Kowloon	1	1	0	0	1	0
Discovery College	1	1	0	0	0	1

School	##	#	G	S	B	HM
HKBU Affiliated School Wong Kam Fai Secondary and Primary School	1	1	0	0	0	1
West Island School	1	1	0	0	0	1
Ha Kwai Chung Government Secondary Technical School	1	1	0	0	0	0
SKH Leung Kwai Yee Secondary School	1	1	0	0	0	0
Tsuen Wan Public Ho Chuen Yiu Memorial College	1	1	0	0	0	0
Tuen Mun Government Secondary School	1	1	0	0	0	0

Here are some records related to the schools that can be deduced from the team list.

The school which has team members in the most number of consecutive years:
Queen Elizabeth School (10 years, from 1997 to 2006)

The schools which have the most number of team members in one IMO:
King's College (1992, 3 team members)
La Salle College (2014, 3 team members)
St. Paul's Co-educational College (1989, 3 team members)

How do our team perform in general? Here is a table listing all the awards obtained by team members of different grades.

Notes:

- Total includes all results since 1988, while all other items only include results since 2000, because some information in the old days cannot be found.
- # denotes the number of team members counted with multiplicity (so far no team member repeats the same grade)
- G, S, B and HM denote the numbers of gold medals, silver medals, bronze medals, and honourable mentions respectively.
- G13 is regarded as S7, G12 is regarded as S6, etc.

Grade	#	G	S	B	HM
Total	216	14 ↑	61 ↑	92 ↑	25
P6	1	0	1	0	0
S1	1	1	0	0	0
S2	2	0	0	2 ↑	0
S3	7	0	0	4 ↑	1

Grade	#	G	S	B	HM
S4	17	0	9	7	0
S5	36	6	15 ↑	13 ↑	2
S6	55	5 ↑	19	21 ↑	8
S7	25	2	8	7	4

In view of this, there is an easy way to get a gold medal in the IMO. Just make sure that you get into the team in S1!

2 Facts in the IMO

Let's learn more about the IMO via this Q&A session.

Notes:

- Some of the answers only refer to the situations in the modern IMO, while some facts in the old days are unknown.

Q. When was the first IMO held?

A. Year 1959. Among all the International Science Olympiads, the IMO has the longest history.

Q. Does the IMO hold every year?

A. Yes, except in 1980. In 1980, the host of the IMO was supposed to be Mongolia. But then later on Mongolia decided not to host the event, and no other countries took up the job. There are different rumours for the reason behind. It is generally believed that this is due to the internal strife of Mongolia at that time.

Q. When and where does the IMO hold each year?

A. Mainly in July. Some IMOs in the old days started in late June. The only exception is IMO 2020. In that year, the IMO was postponed to September due to the uncertainty arisen from the coronavirus.

All countries can be the host of the IMO. Normally, decisions on the host country would be made several years beforehand.

Q. Who are eligible to join the IMO?

A. All contestants must have been born less than 20 years before 1 July in the year of participation at the IMO. Also, they must have been enrolled in full-time primary or secondary education on or after 1 December in the year before, or they are home-schooled students not having received a high-school diploma, or equivalent. Other than that, one can participate in the IMO for any number of times. Apart from this official regulation, different countries may have different policies in their selection process of the team.

Initially, each country can send up to 8 team members. In 1982, each country was only allowed to send 4 team members. Since 1983 onwards, all teams can send up to 6 team members.

Q. What is the format of the competition?

A. The competition is held on 2 consecutive days. The format was not fixed in the old days. For example, there were 7 problems in 1960 and 1962. But other than these, there are exactly 6 problems in every other year, with 3 problems on each day. The contestants can work on the paper for 4.5 hours on each day. Traditionally, P1 and P4 should be the easiest, P2 and P5 are of medium level, while P3 and P6 are the hardest.

Q. Where do the problems in the IMO come from?

A. Each participating country other than the host is expected to submit at most 6 problems some months before the IMO. However, this is not strict. I can recall that some country submitted 10 problems in a certain year. It is said that only the leader of each team may submit problems. But it all depends on the situation of different teams. For us, only the EDB can submit problems, of course under the suggestions of the leader.

In recent years, more than 100 problems are received each year. These are known as the problems in the longlist. Next, the problem selection committee, consisting of around 5 to 10 people appointed by the host organization, selects around 30 problems among these, and make up a shortlist. Several days before the contests, the jury, including all leaders and a chair, discuss the shortlist problems and choose 6 problems for the IMO. However, in 2020 and 2021, the IMO was a virtual event due to the coronavirus. Thus, there was no jury meeting to discuss the shortlist problems, and the problem selection committee was responsible to choose the 6 problems for the contest.

Q. Where can I see the longlist and shortlist problems?

A. Most of the longlist problems from 1966 to 1992 can be found online, such as the contest collections in the AoPS. Afterwards, it was decided that the longlist problems would not be disclosed to any person other than the members in the problem selection committee, so that the unused problems can be further used in different competitions or selection tests.

The shortlist problems are given to the leaders in each IMO, and are kept confidential until the conclusion of the next IMO. All shortlist problems in the past years can be found in the AoPS. But in the official website, only the shortlist problems since 2006 are available.

Q. It seems like some people already know the shortlist problems before the next IMO. How can they find the problems before the shortlist is officially released?

A. The leader is the first one who knows the shortlist problems in each team. They are allowed to use these problems for the purpose of training and team selection.

Therefore, the leaders can actually share the shortlist problems with the other trainers and also the trainees. The only restriction is that every time when the shortlist problems are disclosed to somebody else, they should be notified that this is confidential!

Q. If all leaders of the teams can choose the problems for the IMO, does it mean that every team would know the contest problems beforehand?

A. In the IMO, the leader of each team should be separated from the deputy leader and all contestants, and should not contact them or leak the problems before the contests on both days are finished. In reality, and in view of the current development of the technology, no one can guarantee that there is really no contact between the two groups. But then this is an ethical issue, and we tend to believe that everyone is following the rules.

Q. What will happen if some leaders do not follow the rules and leak the problems before the contest?

A. Different kinds of cheating cases did happen or were believed to have happened before. Before around 2010, there were no concrete policies telling us how to handle the suspected cheating cases. In most situations, there was no concrete evidence that could show that a cheating had occurred for 100% sure. In many cases, the decision and the punishment, usually a disqualification, were made by voting in the jury meeting. However, there are some obvious drawbacks of this, as different leaders may have bias on some other teams or countries, and the jury only have a short period of time to learn the details because the decision has to be made before the final jury meeting where the results are released. It is possible that the jury only discuss the issue for less than one hour before making the decision.

In 2010, there was a heated discussion about the disqualification of one team. Since then, the jury decided to set up an ethics committee consisting of a few members selected by the jury to handle related cases, following some more systematic guidelines. One famous irregularity case handled by this committee happened in 2018. Although at the end it was still the jury to vote for the final decision, the ethics committee could put more effort in investigating the case, and showed all the evidences to the leaders before they voted.

Therefore, just like all kinds of competitions, it is impossible to prevent cheating. But there are some reliable systems to handle such cases if they happen.

Apart from these, it happened in 2015 that the day 2 problems were mistakenly given to some deputy leaders on day 1. The jury reacted quickly and set a new paper for day 2 in the evening of day 1, and translated all the paper at night. Luckily, the contest was still held smoothly, except that the problems turned out to be harder than usual.

Q. Can you say more about the details of the cheating cases in 2010 and 2018?

A. I was not involved in any jury meetings of these years, and even if I were, probably I couldn't tell anything concrete, because the details should be kept confidential as a protection of everyone involved. In the following, I could only mention some rumours found online. Clearly, these are not reliable, and you should treat them as some gossip.

In 2010, a whole team was disqualified, which was formally announced in the official website. The main evidence was that most, some said 4, of the team members stated the same crucial lemma when solving a hard problem P3 which was also used in the official solution. It was not known whether all of them eventually solved this problem, and how did the remaining team members performed in this problem. There were no similarities found in other problems. The contestants used their own languages in their country to write down the solutions, and the whole team including the leader could not communicate in English efficiently. This made it even harder to make the judgement.

In view of the information known to the public, it seemed to be a hard decision to disqualify the whole team. Indeed, that country is one of the strong teams in the IMO, and could get into the top 10 for many years. It is not too surprising if some team members could solve the same problem in a similar manner. There might also be bias on this team because they had been caught to cheat in 1991 as well with strong evidence. But that was already 20 years before. Also, some said it was too harsh to punish the whole team if not all team members had cheated. Anyway, since we do not know the details that the jury would know, we could never tell whether the decision was correct or not. And it was exactly the reason why the ethics committee was set up after this debatable issue.

In 2018, only 3 contestants involved instead of the whole team were disqualified. The information cannot be found in the official website, as we simply regard them as not participating in this competition. Similar to the previous case, 3 contestants from the same team were found to present a very similar solution to a hard problem P6, which couldn't be found in the complete solutions from any other contestants or the official solution. The contestants claimed the same trick was used in a particular problem in the old IMO shortlist, and they had seen this in the training. But it was also said that only the first idea used in that problem was similar and many other solutions shared the same first step, while the remaining works of the 3 suspected scripts were highly similar as well. Again, all these claims are not well-justified.

Anyway, just like what the IMO Oath says, we must respect and abide by the rules govern it, and commit ourselves to honest conduct in the true spirit of fair play, for the glory of mathematics and the honour of our teams. Don't attempt to cheat in any case!

Q. What is the scope of the IMO?

- A. There is no scope written down officially. The problems may cover various topics of pre-university mathematics, but this description is vague. For example, some may think that calculus should not be included, but actually it appears occasionally in the IMO. Three-dimensional geometry used to be a common topic. But the last time when such a problem appears in the IMO was 1979. In 1999, there was a problem in the shortlist involving three-dimensional geometry which was selected as an IMO problem. However, the problem statement changed in the final version of the paper so that one was restricted to consider plane geometry only.

Since 1993, the shortlist problems are classified into 4 topics, namely, algebra, combinatorics, geometry, and number theory. In 1995, there was an extra topic called sequence. In 1997, the classification is unknown.

Q. Is it true that there are problems from all the 4 topics in each IMO?

- A. Traditionally, there should be at least one and at most two problems from each topic in each IMO, so that the paper is more balanced. The only exceptions are 1993 and 1994. In 1993, there were 3 geometry problems but no number theory. In 1994, there were 3 number theory problems but no combinatorics.

In 2011, the problem distribution was awkward. According to the classification in the shortlist, there were two algebra problems on day 1. However, P1 involves more number theory instead. As a result, there were two number theory problems P1 and P5, and two combinatorics problems P2 and P4, and so there were no easy algebra and geometry problems. In view of this, Geoff Smith from the United Kingdom, suggested using a new protocol to select the problems.

The new protocol was implemented in 2013, and it was used in every year since then. The protocol says that there should be one problem from each topic among P1, P2, P4 and P5, while there is no restriction on the hard problems except that they should not be of the same topic. However, even under this protocol, sometimes the real paper may not be balanced due to the distribution of problems in the shortlist. For example, it is well-known that there were 3 combinatorics problems but no number theory in 2014, because N3 is a combinatorial problem put on the list of number theory.

Q. When and how are the scripts marked?

- A. After the competitions on day 1 and day 2, all the scripts are scanned, and the leader can get back the hard copies. There are coordinators appointed by the host organization who will read the scripts immediately after the contest. Each coordinator only focuses on one problem, and there are around 10 to 20 coordinators who mark the same problem, including a problem captain who is

in charge of that problem. Normally, each script is read by 2 coordinators, and they will discuss and propose a score to each script which is not known to the teams. At the same time, the leader and the deputy leader, possibly together with some observers, will read the scripts of their own team, and propose another score for each script. Note that there is a marking scheme known to both the coordinators and the leaders when they make the judgements. In the old days, each problem was worth 8 or 9 points. Since 1979 till now, each problem is worth 7 points, regardless of their difficulties.

Then on day 3 and in the morning of day 4, there are some scheduled meetings where the leader and the deputy leader, perhaps also an observer, of each team will meet the 2 coordinators who mark the scripts of their team. Usually, the leader first proposes the scores of their team members in that problem and explain. If these match the decisions of the coordinators, the scores are confirmed and the two parties sign on a form. Normally, no change can be made afterwards. If there are discrepancies, the two parties will further discuss and elaborate their thoughts. If there is no conclusion when the meeting ends, they will schedule an additional meeting afterwards. In many cases, the problem captain may eventually be involved and join the discussion. If the issue is still not settled down, the case will be passed to the final jury meeting, where the leader and the coordinator have to explain to other leaders, and let the jury make the final decision by voting.

- Q. If there is a marking scheme, why would there be discrepancies between the coordinators and the leaders? Is that the leaders do not follow the marking scheme and just ask for more and more points?
- A. In the instruction of coordination, it is said that the leaders should act responsibly and do not ask for fantasy marks. Most leaders should be following this, as this affects the reputation of their countries. Of course there are still some funny things which may happen occasionally. For example, it is quite common that a leader requests for 1 point in a problem because of the hard work of the contestant, and because they do not want to get 0 points for the whole team in that problem.

Indeed, the main reason why there are many debates during the coordination process is that the IMO is very different from a school exam. There are many different approaches to solve a problem and different ways to present the ideas. It is impossible for a marking scheme to cover all situations because it is made before the contest. Therefore, the first principle of the marking scheme is that, any equivalent approach should receive proper and proportionately judged equivalent marks. This gives some room for the leaders to ask for more partial credit. Even for a complete solution, there may be arguments whether some statements are obvious or not, and whether the contestant has provided all sufficient details. In some teams, the team members might use a language that is not understood

by any of the coordinators. So a task of the leader is to translate the script and explain every detail to the coordinators during the meeting.

In addition, since there is only a short period of time for the coordinators to read a lot of scripts, they may easily miss something. It is especially true if a contestant submits dozens of rough work paper. It has happened that some contestants submitted over a hundred pieces of paper in a single problem! Indeed, one principle of the marking is that whenever a contestant has written down something that deserves some points, we should award the points, no matter how many other wrong things are written down, or whether this is put on the answer sheet or the rough work paper. Clearly, the coordinators do not bother to check every single page carefully. Therefore, one major task of the leader and the deputy leader is to search for every single place in the submitted scripts which may deserve some points. Very often, the things written down on the rough work paper are vague, and so there will be arguments on those issues.

Q. When are the results released?

A. Once the leader confirms the scores with the coordinators, they already know the results of their team. On day 3 and day 4, usually there are some screens showing the confirmed scores of all teams that everyone can view, which keep on updating. Whenever a team finishes the coordination in one problem, the screens will show the scores of their team members, except one of them which is hidden by a black dot. For a full team of 6 members, each team member is assigned a distinct problem randomly so that the score in that problem is hidden. This allows some mystery of the scores. At night of day 4, the final jury meeting will be held, and all results will be available on the official website right after that.

Q. What awards can one get in the IMO?

A. Initially, there were only gold medals, silver medals, and bronze medals. Of course the best group of contestants would get gold medals, the second best group would get silver medals, and the next best group would get bronze medals. In 1988, exactly the year that the Hong Kong team first joined the IMO, they started to award honourable mentions to those contestants who could not get any medals but could solve at least one problem perfectly.

Apart from the medals awarded for the outstanding performances, there are some special prizes awarded to contestants who provide a nice solution to one of the problems. Special prizes were mainly awarded in the old days. The last special prize was awarded in 2005, and the second last was awarded in 1995.

In addition, there is an unofficial award called the IMO Maryam Mirzakhani award given since 2017, initially called the Olympic Girls' IMPA award. This award is given to the female contestant who performs the best in each continent.

At the moment the awards were only made by the host organization each year, and have not been formally included in the IMO regulation.

Q. How many contestants can get gold medals, silver medals, and bronze medals?

A. In the IMO regulation, it is stated that the numbers of gold medals, silver medals, and bronze medals must be approximately in the ratio $1 : 2 : 3$, and the total number of awards must not exceed half the total number of contestants, unless it is approved by at least two thirds of the members of the jury. Of course contestants obtaining the same score must be given the same prize. Normally, there are 2 ways of setting the cutting scores which are close to the ideal case. The jury vote for one of these without knowing the exact cutting scores but only the numbers of contestants obtaining the medals in different cases.

Q. Can you show me some outstanding solutions that were awarded the special prizes?

A. Sure. The first example is the special prize awarded to Iurie Boreico in 2005 for the following problem.

Let x, y, z be three positive reals such that $xyz \geq 1$. Prove that

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0.$$

Proof. We have

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} - \frac{x^5 - x^2}{x^3(x^2 + y^2 + z^2)} = \frac{x^2(x^3 - 1)^2(y^2 + z^2)}{x^3(x^5 + y^2 + z^2)(x^2 + y^2 + z^2)} \geq 0.$$

So

$$\begin{aligned} \sum_{\text{cyc}} \frac{x^5 - x^2}{x^5 + y^2 + z^2} &\geq \sum_{\text{cyc}} \frac{x^5 - x^2}{x^3(x^2 + y^2 + z^2)} = \frac{1}{x^2 + y^2 + z^2} \sum_{\text{cyc}} \left(x^2 - \frac{1}{x} \right) \\ &\geq \frac{1}{x^2 + y^2 + z^2} \sum_{\text{cyc}} (x^2 - yz) \geq 0 \end{aligned}$$

as $xyz \geq 1$. □

The second example is the special prize awarded to Nikolay Nikolov in 1995 for the following problem.

Let p be an odd prime number. How many p -element subsets A of $\{1, 2, \dots, 2p\}$ are there, the sum of whose elements is divisible by p ?

Proof. Let ω be a p th root of unity. Then we have

$$\prod_{k=1}^{2p} (x - \omega^k) = (x^p - 1)^2 = x^{2p} - 2x^p + 1.$$

Comparing the coefficients of x^p on both sides, we obtain

$$\sum_{1 \leq j_1 < j_2 < \dots < j_p \leq 2p} \omega^{j_1} \omega^{j_2} \dots \omega^{j_p} = 2.$$

Note that the sum on the left is equal to $\sum a_j \omega^j$, where a_j represents the number of subsets $\{j_1, j_2, \dots, j_p\}$ of $\{1, 2, \dots, 2p\}$ such that $j_1 + j_2 + \dots + j_p \equiv j \pmod{p}$. This implies

$$(a_0 - 2) + a_1 \omega + a_2 \omega^2 + \dots + a_{p-1} \omega^{p-1} = 0.$$

Since the minimal polynomial of ω over $\mathbb{Q}$ is $1 + x + x^2 + \dots + x^{p-1}$, it follows that $a_0 - 2 = a_1 = a_2 = \dots = a_{p-1}$. As we also have $a_0 + a_1 + a_2 + \dots + a_{p-1} = \binom{2p}{p}$, we easily obtain

$$a_0 = \frac{1}{p} \left[\binom{2p}{p} - 2 \right] + 2.$$

□

Another example is the well-known method of Vieta jumping in solving 1988 P6.

3 Statistics of the IMO

All the results in the IMO except those in the old days are available on the official website. This section lists out some interesting records deduced from these.

Country Results

As usual, ‘country’ refers to a country or in some exceptional cases to a territory/geographical region.

Number of countries which have registered in the IMO: 142

Number of countries which have participated in at least one IMO: 138

Among these, 7 of the countries no longer exist.

The following table provides some basic information of all participating countries.

Notes:

- Year denotes the first year of participation.
- BS denotes the best total score since 1983 (when the maximum total score is fixed at $42 \times 6 = 252$), while BR denotes the best rank since 1959. The numbers in the brackets refer to the best total score and the best rank since 2000, if they are different from the overall results. The symbol - means the country does not participate in the IMO.
- G, S, B and HM denote the numbers of gold medals, silver medals, bronze medals, and honourable mentions respectively.

Code	Country	Year	BS	BR	G	S	B	HM
CHN	China	1985	252	1	180	36	6	0
USA	United States	1974	252(227)	1	146	119	30	1
RUS	Russia	1992	227(215)	1	106	62	12	0
KOR	South Korea	1988	226	1	93	79	28	7
HUN	Hungary	1959	221(187)	1(7)	86	171	115	10
ROU	Romania	1959	250(208)	1(4)	85	154	110	7
USS	Soviet Union	1959	241(-)	1(-)	77	67	45	0
VNM	Vietnam	1974	220(196)	3	69	115	82	2
BGR	Bulgaria	1959	227	1	57	127	119	14
UNK	United Kingdom ↑	1967	206(179)	2(7)	54	121	130	18
GER	Germany	1977	248(192)	1(4)	54	112	87	16
IRN	Iran	1985	217(201)	1(4)	50	110	49	4

Code	Country	Year	BS	BR	G	S	B	HM
JPN	Japan	1990	212	2	46	96	51	5
TWN	Taiwan	1992	192	3	44	103	36	8
UKR	Ukraine	1992	195(186)	4	43	74	60	9
CAN	Canada	1981	183	5	42	67	96	20
THA	Thailand ↑	1989	193	5	34	67	52	24
POL	Poland	1959	183	2(6)	33	94	143	28
AUS	Australia	1981	187(169)	6	28	77	100	19
FRA	France	1967	175(158)	5(14)	26	69	128	28
GDR	German Democratic Republic	1959	231(-)	1(-)	26	62	60	0
SGP	Singapore	1988	196	3	24	69	74	23
PRK	North Korea	1990	187	4	22	36	9	2
ITA	Italy ↑	1967	188	6	21	51	78	35
TUR	Turkey	1978	177	6	20	72	93	15
ISR	Israel	1979	188	7	20	66	105	23
IND	India ↑	1989	187(174)	7	16	73	79	28
KAZ	Kazakhstan	1992	174	5	16	44	79	29
SRB	Serbia	2006	171	9	16	33	38	17
BLR	Belarus	1992	167	6(7)	15	61	84	18
HKG	Hong Kong	1988	173	9	14	61	92	25
BRA	Brazil	1979	173	10	14	55	89	35
AUT	Austria	1970	150(118)	4(36)	13	35	112	63
NLD	Netherlands	1969	155	5(13)	11	34	91	51
CZS	Czechoslovakia	1959	202(-)	1(-)	10	50	73	2
CZE	Czechia	1993	154(145)	10(14)	7	37	78	43
SWE	Sweden	1967	138	6(24)	7	34	86	64
MNG	Mongolia ↑	1964	162	8(11)	7	30	78	57
YUG	Yugoslavia	1963	170(114)	4(23)	6	46	96	7
HRV	Croatia	1993	157	15	6	27	85	40
ARG	Argentina	1988	159(139)	20(22)	6	27	69	50
MAS	Malaysia ↑	1995	137	23	6	16	31	43
PER	Peru	1987	173	16	5	42	60	33
SVK	Slovakia	1993	150(140)	12(18)	5	41	86	36
MEX	Mexico ↑	1981	167	14	5	33	78	40
IDN	Indonesia	1988	171	10	5	30	58	35
MDA	Moldova	1993	140	9	5	24	58	51
GEO	Georgia	1993	149	12	5	23	78	55
HEL	Greece ↑	1975	163	12	4	31	81	63
PHI	Philippines	1988	145	17	4	19	40	30
NOR	Norway	1984	122	18(34)	3	15	40	50
POR	Portugal	1989	137	33	3	8	41	46
SUI	Switzerland ↑	1991	145	24(30)	2	13	62	44

Code	Country	Year	BS	BR	G	S	B	HM
NZL	New Zealand	1988	116(114)	29	2	12	59	67
MAC	Macau	1990	127	35	2	5	36	62
LUX	Luxembourg	1979	65	23(53)	2	5	23	25
CIS	Commonwealth of Independent States	1992	176(-)	4(-)	2	3	0	1
ARM	Armenia	1992	163	20	1	30	71	45
COL	Colombia	1981	136(129)	21(27)	1	19	80	53
BEL	Belgium	1969	121(99)	13(37)	1	18	72	73
LVA	Latvia	1992	124(101)	22(38)	1	12	44	69
FIN	Finland	1965	103(101)	8(38)	1	11	56	72
LTU	Lithuania	1992	129	31	1	10	54	64
SAF	South Africa	1992	120(110)	27	1	10	53	77
MKD	North Macedonia ↑	1993	148	31(36)	1	9	50	48
SVN	Slovenia	1993	117	34	1	7	41	73
CUB	Cuba	1971	92	14(26)	1	7	37	28
BGD	Bangladesh	2005	118	26	1	7	35	40
DEN	Denmark	1991	121	32(44)	1	6	41	57
TUN	Tunisia	1981	85	19(56)	1	6	18	45
PRI	Puerto Rico	1997	55	69	1	1	7	18
SAU	Saudi Arabia	2004	168	22	0	11	41	18
BIH	Bosnia ↑	1993	154	29	0	10	61	61
AZE	Azerbaijan ↑	1992	157	28	0	10	44	63
EST	Estonia	1992	124	36	0	10	37	67
UZB	Uzbekistan	1997	124	26	0	10	32	50
ESP	Spain	1983	139	22(31)	0	8	64	62
TJK	Tajikistan	2005	107	42	0	5	26	54
CYP	Cyprus ↑	1984	103	26(34)	0	5	21	58
SYR	Syria	2009	102	47	0	5	13	39
SCG	Serbia and Montenegro	2003	132	21	0	5	7	3
MAR	Morocco	1983	138(88)	18(41)	0	4	36	83
TKM	Turkmenistan	1993	97	34	0	4	33	48
CHI	Chile	1994	67	54(63)	0	3	6	30
MNE	Montenegro	2007	50	73	0	3	2	18
IRL	Ireland	1988	84	40(51)	0	2	11	57
ALB	Albania	1993	76	50	0	2	8	42
ALG	Algeria	1977	100	21(57)	0	2	8	20
TTO	Trinidad and Tobago	1991	63(40)	45(59)	0	2	5	29
VEN	Venezuela	1981	59	24(49)	0	2	5	27
CRI	Costa Rica	2005	92	47	0	1	20	46

Code	Country	Year	BS	BR	G	S	B	HM
KGZ	Kyrgyzstan	1993	79	49	0	1	12	55
ISL	Iceland	1985	63	32(59)	0	1	12	47
PAR	Paraguay	1997	59	67	0	1	11	27
PAK	Pakistan	2005	58	73	0	1	8	23
SLV	El Salvador	2005	61	68	0	1	6	34
LIE	Liechtenstein	2005	22	84	0	1	4	6
CIV	Ivory Coast	2010	54	69	0	1	0	8
LKA	Sri Lanka	1995	82	50	0	0	24	63
ECU	Ecuador	1988	71	49(64)	0	0	11	37
PAN	Panama	1987	42	42(72)	0	0	5	19
NGA	Nigeria	2006	52	67	0	0	5	16
NIC	Nicaragua	1987	45	41(80)	0	0	3	18
URY	Uruguay	1987	47	38(65)	0	0	2	31
KSV	Kosovo	2011	55	77	0	0	2	26
HND	Honduras	2008	31	81	0	0	1	11
GTM	Guatemala	1997	20	75(76)	0	0	1	6
IRQ	Iraq	2016	17	89	0	0	1	3
UAE	United Arab Emirates ↑	2008	36	84	0	0	1	2
GHA	Ghana	2014	21	93	0	0	1	2
KWT	Kuwait	1982	53(12)	30(78)	0	0	1	1
OMN	Oman ↑	2020	27	90	0	0	1	0
BOL	Bolivia	1997	57	74(77)	0	0	0	20
KHM	Cambodia	2007	58	66	0	0	0	14
MMR	Myanmar	2016	32	85	0	0	0	8
UGA	Uganda	2012	29	87	0	0	0	6
MRT	Mauritania	2009	42	84	0	0	0	5
NPL	Nepal	2017	27	89	0	0	0	5
RWA	Rwanda ↑	2021	35	85	0	0	0	4
BWA	Botswana	2015	19	95	0	0	0	4
DOM	Dominican Republic	2019	26	93	0	0	0	2
BAH	Bahrain	1990	65(-)	40(-)	0	0	0	1
BFA	Burkina Faso	2014	19	89	0	0	0	1
TZA	Tanzania	2014	14	98	0	0	0	1
PSE	Palestine	2022	14	99	0	0	0	1
EGY	Egypt	2016	12	96	0	0	0	1
NCY	Northern Cyprus	1993	17(-)	67(-)	0	0	0	0
MOZ	Mozambique	2004	13	82	0	0	0	0
KEN	Kenya	2016	11	99	0	0	0	0
MDG	Madagascar	2016	10	100	0	0	0	0
JAM	Jamaica	2016	9	102	0	0	0	0
BRU	Brunei	2000	8	81	0	0	0	0

Code	Country	Year	BS	BR	G	S	B	HM
CMR	Cameroon	2023	6	109	0	0	0	0
ZWE	Zimbabwe	2009	5	94	0	0	0	0
BEN	Benin	2009	3	98	0	0	0	0
GMB	Gambia	2014	1	100	0	0	0	0
LAO	Laos	2016	0	109	0	0	0	0
AGO	Angola	2019	0	110	0	0	0	0
AFG	Afghanistan	-	-	-	-	-	-	-
BTN	Bhutan	-	-	-	-	-	-	-
SEN	Senegal	-	-	-	-	-	-	-
YEM	Yemen	-	-	-	-	-	-	-

Countries which have participated in all the IMOs: BGR, ROU

Let's see some remarkable results of the countries.

Countries which have obtained the first rank in one IMO:

CHN (×24), USS (×14), USA (×8), HUN (×6), ROU (×5), GER (×2), KOR (×2),
RUS (×2), BGR (×1), CZS (×1), GDR (×1), IRN (×1)

Country which has obtained the first rank in the most number of consecutive years:
USS (1963—1967, 5 consecutive years), CHN (2019—2023, 5 consecutive years)

Countries with the highest total scores since 1983:

USA (1994, 252 points), CHN (2022, 252 points)
ROU (1987, 250 points)
GER (1987, 248 points)
USS (1991, 241 points)
CHN (1992, 240 points), CHN (2023, 240 points)

Countries with the highest total scores since 2000:

CHN (2022, 252 points)
CHN (2023, 240 points)
CHN (2005, 235 points)
BGR (2003, 227 points), CHN (2019, 227 points), USA (2019, 227 points)
KOR (2019, 226 points)

Countries which have perfect scorers since 2000:

×29: CHN
×10: USA
×7: RUS
×5: KOR
×4: JPN
×3: TWN, VNM
×2: BLR, CAN, GER, MDA, UKR
×1: AUS, BGR, HUN, IRN, POL, ROU, SGP, UNK

Countries which have obtained at least 2 golds in one IMO since 1983:

AUS, BLR, BRA, BGR, CAN, CHN, CIS, HRV, CZS, FRA, GDR, GER, HKG, HUN, IND, IDN, IRN, ISR, ITA, JPN, KAZ, PRK, KOR, MAS, MDA, MNG, NLD, PER, PHI, POL, ROU, RUS, SRB, SGP, TWN, THA, TUR, UKR, UNK, USA, USS, VNM

Countries which have obtained at least 3 golds in one IMO since 1983:

BLR, BGR, CAN, CHN, FRA, GDR, GER, HKG, HUN, IND, IRN, ISR, JPN, KAZ, PRK, KOR, NLD, ROU, RUS, SRB, SGP, TWN, THA, TUR, UKR, UNK, USA, USS, VNM

Countries which have obtained at least 4 golds in one IMO since 1983:

BGR, CHN, GER, HUN, IRN, JPN, KAZ, KOR, ROU, RUS, SGP, TWN, UKR, USA, USS, VNM

Countries which have obtained at least 5 golds in one IMO since 1983:

BGR, CHN, IRN, JPN, KOR, ROU, RUS, USA, USS

Countries which have obtained 6 golds in one IMO since 1983:

BGR, CHN, KOR, RUS, USA

The following are some special results.

Countries which have obtained 6 silvers in one IMO since 1983:

GER, JPN, RUS, UKR

Countries which have obtained 6 bronzes in one IMO since 1983:

AUS, BLR, BRA, COL, PER, VNM

Countries which have obtained 6 HMs in one IMO since 1988:

BEL, FIN, IRL, KGZ, LVA, SAF, TJK, TUN, UZB

Country with the best improvement in the total score compared to the previous IMO since 1983: CHN, 150 points (from 27 points in 1985 to 177 points in 1986)

Country with the best improvement in the total score compared to the previous IMO since 2000: LTU, 98 points (from 31 points in 2021 to 129 points in 2022)

Country with the best improvement in the rank compared to the previous IMO since 1983: GEO, 47 places (from rank 59 in 2016 to rank 12 in 2017)

Countries with the longest sequence of strictly increasing scores in consecutive years:
CYP (2005–2010, 14, 19, 41, 42, 45, 62 points) (2012–2017, 39, 52, 53, 58, 65, 93 points)
ECU (2001–2006, 0, 3, 11, 14, 17, 40 points)
FIN (2015–2020, 26, 55, 56, 70, 78, 81 points)
MAS (2009–2014, 31, 75, 93, 100, 117, 129 points)
POR (2001–2006, 6, 15, 22, 26, 27, 78 points)
SVN (2015–2020, 46, 65, 90, 104, 109, 117 points)

Country with the longest sequence of strictly decreasing scores in consecutive years:
UNK (1994–2001, 206, 180, 161, 144, 122, 100, 96, 79 points)

Countries with the longest sequence of strictly increasing ranks in consecutive years:
AUS (1997–2002, ranks 9, 13, 15, 16, 25, 26)
PHI (1991–1996, ranks 42, 44, 52, 53, 65, 74)
UNK (1996–2001, ranks 5, 16, 17, 20, 22, 31)
IDN (2018–2023, ranks 10, 14, 32, 33, 38, 39)

Countries with the longest sequence of monotonic increasing ranks in consecutive years:
KWT (1993–2002, ranks 68, 68, 73, 75, 76, 76, 78, 78, 81, 83)
ROU (1959–1968, ranks 1, 2, 3, 3, 3, 3, 3, 5, 5, 8)

Countries with the longest sequence of strictly decreasing ranks in consecutive years:
GDR (1963–1968, ranks 7, 6, 5, 3, 2, 1)
MAS (2009–2014, ranks 75, 54, 41, 35, 31, 23)
SYR (2011–2016, ranks 92, 89, 88, 72, 54, 48)

Country with the longest sequence of monotonic decreasing ranks in consecutive years: TUR (2004–2011, ranks 32, 21, 19, 17, 8, 8, 8, 6)

Country with the same score, same rank and the same medals in 2 consecutive years:
BOL (2013–2014, 5 points, rank 94, 0 G, 0 S, 0 B, 0 HM)

Country with the same score in 3 consecutive years: BOL (2013–2015, 5 points)

Maximum number of countries with the same score in one IMO: 5
BGD, KWT, LUX, SAU, TJK (2005, 3 points, rank 85)
ARG, LTU, MDA, SVK, SUI (2010, 92 points, rank 39)

How do the countries perform in some individual problems?

Countries which have got perfect scores in at least 2 problems in one IMO since 2000:

ARM, AUS, BGD, BLR, BIH, BRA, BGR, CAN, CHN, HRV, FRA, GEO, GER, HEL, HKG, HUN, IDN, IRN, ISR, ITA, JPN, KAZ, PRK, KOR, MEX, MNG, NLD, NZL, MKD, PER, PHI, POL, ROU, RUS, SAU, SGP, **SVK**, SWE, TWN, THA, TUR, UKR, UNK, USA, VNM

Countries which have got perfect scores in at least 3 problems in one IMO since 2000:

ARM, BGR, CHN, GER, IRN, JPN, KOR, POL, ROU, RUS, SGP, TWN, THA, **TUR**, UKR, UNK, USA, **VNM**

Countries which have got perfect scores in at least 4 problems in one IMO since 2000:

CHN, **ROU**, RUS, **USA**

Countries which have got perfect scores in at least 5 problems in one IMO since 2000:

CHN

Countries which have been the only team to perform the best in one problem since 2000:

AUS (2017 P3, 8 points)

BGR (2003 P3, 41 points) (2003 P6, 36 points) (2008 P6, 16 points)

CHN (2000 P6, 34 points) (2001 P5, 42 points) (2002 P3, 24 points)

(2003 P2, 40 points) (2003 P5, 42 points) (2004 P6, 42 points)

(2005 P6, 42 points) (2006 P2, 42 points) (2006 P3, 35 points)

(2006 P5, 38 points) (2006 P6, 16 points) (2007 P3, 17 points)

(2007 P5, 42 points) (2008 P3, 42 points) (2009 P3, 42 points)

(2010 P6, 25 points) (2011 P3, 42 points) (2012 P6, 30 points)

(2014 P6, 24 points) (2015 P2, 36 points) (2015 P6, 27 points)

(2016 P6, 28 points) (2017 P6, 31 points) (2018 P3, 17 points)

(2019 P6, 36 points) (2000 P2, 38 points) (2000 P3, 31 points)

(2000 P6, 20 points) (2021 P6, 42 points) (2022 P3, 42 points)

(2022 P6, 42 points) **(2023 P5, 42 points)** **(2023 P6, 30 points)**

HKG (2017 P5, 26 points)

IRN (2013 P3, 31 points)

ISR (2021 P2, 22 points)

JPN (2009 P6, 19 points) (2014 P5, 37 points)

KOR (2010 P3, 24 points) (2012 P5, 42 points) (2013 P6, 20 points)

(2017 P2, 39 points) (2019 P3, 31 points)

RUS (2004 P3, 25 points) (2007 P6, 9 points) (2014 P3, 29 points)

SGP (2011 P2, 27 points)

TWN (2000 P4, 40 points)

TUR (2011 P6, 14 points) (2012 P4, 42 points)

USA (2000 P3, 27 points) (2001 P3, 28 points) (2002 P6, 31 points)

(2012 P3, 33 points) (2016 P3, 27 points) (2018 P6, 31 points)

VNM (2004 P5, 38 points)

In each IMO, there is a host country to organize the event. It is said that the host country usually performs better. Here are some reasons. Firstly, the team of the host country don't need to go on a long flight some days before the exam days, and they don't need to adapt to jet lag. Also, it is usually more comfortable to work at their own places. Another common reason is that some countries may have put more effort in training a group of team members since several years before they host the IMO so as to show their best to the rest of the world. Let's see some evidence in the IMO history.

Countries which have been the host in one IMO:

×6: ROU

×3: CZS, HUN, POL, RUS, UNK, USS

×2: ARG, BGR, GDR, GER, HKG, JPN, USA, YUG

×1: AUS, AUT, BRA, CAN, CHN, COL, CUB, FIN, FRA, HEL, IND, KAZ, KOR, MEX, NLD, NOR, SVN, SAF, ESP, SWE, TWN, THA, TUR, VNM

Remarkable results of host countries:

1959: (ROU) obtaining rank 1 for the first time

1961: (HUN) obtaining rank 1 for the first time

1964: (USS) obtaining rank 1

1968: (USS) obtaining the highest score 298 points

1970: (HUN) obtaining rank 1

1973: (USS) obtaining rank 1

1976: (AUT) obtaining the best rank 5 (beaten by rank 4 in 1981)

1978: (ROU) obtaining 2 gold medals for the first time, and obtaining rank 1

1981: (USA) obtaining 4 gold medals for the first time, and obtaining rank 1

1988: (AUS) obtaining the second gold medal

1990: (CHN) obtaining 5 gold medals for the first time, and obtaining rank 1

1992: (RUS) first participation

1993: (TUR) obtaining the second silver medal

1994: (HKG) obtaining the highest score 162 points

1996: (IND) obtaining the second gold medal

1998: (TWN) obtaining 3 gold medals for the first time, obtaining the highest score 184 points and the best rank 5 (beaten by 192 points and rank 3 in 2014)

1999: (ROU) obtaining 3 gold medals, and obtaining the best rank 4 since 1999

2000: (KOR) obtaining 3 gold medals, and obtaining the best rank 4 (beaten by rank 1 in 2012, 2017)

2004: (HEL) obtaining the highest score 126 points (beaten by 127 points in 2017)

2005: (MEX) obtaining the best rank 31 (beaten by rank 17 in 2013)

2006: (SVN) obtaining the third silver medal, and obtaining the highest score 90 points (beaten by 117 points in 2020)

2007: (VNM) obtaining 3 gold medals, and obtaining the best rank 3
 2008: (ESP) obtaining the highest score 82 points and the best rank 43 since 2000 (beaten by 133 points and rank 31 in 2020)
 2009: (GER) obtaining the highest score 171 points since 2000
 2010: (KAZ) obtaining 3 gold medals, and obtaining the best rank 5
 2011: (NLD) obtaining the highest score 115 points and the best rank 28 since 2000 (beaten by 155 points and rank 13 in 2014)
 2016: (HKG) obtaining 3 gold medals for the first time, and obtaining the best rank 9
 2021: (RUS) obtaining 5 gold medals

And here is a funny thing found on the official website.

Country which has an anonymous team member: UNK (2005)

2005	?	3	7	7	7	1	7	32	47	91.02%	Silver medal
2005	Saul Glasman	2	7	0	7	3	1	20	139	73.05%	Bronze medal
2005	Nathan Kettle	7	7	7	7	3	2	33	46	91.21%	Silver medal
2005	Matthew Lee	1	7	0	7	7	1	23	114	77.93%	Silver medal
2005	Martin Orr	7	7	6	7	7	1	35	35	93.36%	Gold medal
2005	Jack Shotton	2	7	0	7	0	0	16	191	62.89%	Bronze medal

In fact, it is quite common that some information was lost in the old days.

Individual Results

Here are some statistics related to the individual results in the IMO.

The highest cutting score for gold medals since 1983: 42 points (1987)

The highest cutting score for gold medals since 2000: 35 points (2005)

The lowest cutting score for gold medals since 1983: 24 points (2021)

The most number of contestants obtaining gold medals since 1983: 54 (2011, 2023)

The highest cutting score for silver medals since 1983: 32 points (1987)

The highest cutting score for silver medals since 2000: 29 points (2022)

The lowest cutting score for silver medals since 1983: 19 points (1999, 2003, 2006, 2015, 2017, 2021)

The most number of contestants obtaining silver medals since 1983: 113 (2014)

The highest cutting score for bronze medals since 1983: 23 points (2022)

The lowest cutting score for bronze medals since 1983: 11 points (1993, 2000, 2001)

The most number of contestants obtaining bronze medals since 1983: 170 (2023)

The most number of contestants obtaining HMs since 1988: 222 (2017)

The highest score without medals or HM since 1988: 18 points

1994 EST1 ($6 + 0 + 5 + 2 + 5 + 0 = 18$)

1994 LTU1 ($5 + 2 + 6 + 2 + 1 + 2 = 18$)

The lowest score without 0 points since 1983: 6 points

1990 MEX3 ($1 + 1 + 1 + 1 + 1 + 1 = 6$)

The most number of contestants obtaining perfect scores since 1983: 22 (1987)

The most number of contestants obtaining perfect scores since 2000: 16 (2005)

The most number of contestants obtaining 0 points since 1983: 87 (2021)

The years with no contestants obtaining perfect scores since 1983:

1999 (highest: 39 points), 2007 (highest: 37 points), 2013 (highest: 41 points),
2017 (highest: 35 points)

The years with no contestants obtaining 0 points since 1983:

1990 (lowest: 1 point), 1994 (lowest: 1 point)

The years with exactly 1 contestant obtaining the first rank since 1983:

1996 (ROU5), 1998 (IRN3), 2007 (RUS4), 2010 (CHN3), 2011 (GER4), 2012 (SGP5),
2015 (CAN3), 2020 (CHN2), 2021 (CHN4)

The years with exactly 1 contestant obtaining the last rank since 1983:

1986, 1988, 1990, 1999

Who is the youngest medalist in the IMO? It is quite a well-known fact, although the result is unofficial.

Youngest bronze medalist in the IMO: 1986 AUS6 (10 years old)

Youngest silver medalist in the IMO: 1987 AUS5 (11 years old)

Youngest gold medalist in the IMO: 1988 AUS6 (13 years old)

Youngest perfect scorer in the IMO: 1972 GDR6 (13 years old)

The following provide some funny scores of some contestants.

The highest score without solving any problem:

$6 + 6 + 1 + 4 + 6 + 2 = 25$ points (2002 IRN6)

Gold medalists solving the least number of problems:

1990 ITA2 ($6 + 6 + 5 + 7 + 6 + 6 = 36$ points)

1993 CHN1 ($6 + 7 + 6 + 4 + 5 + 3 = 31$ points)

1996 IND1 ($6 + 1 + 5 + 7 + 5 + 6 = 30$ points)

2012 USA2 ($7 + 5 + 3 + 5 + 4 + 4 = 28$ points)

High scorers without solving P1 and P4:

2019 CHN3 ($6 + 7 + 1 + 6 + 7 + 7 = 34$ points, gold medal)

1989 POL3 ($5 + 4 + 7 + 5 + 7 + 7 = 35$ points, silver medal)

Scores forming a permutation of 2 to 7: $7 + 2 + 4 + 5 + 6 + 3 = 27$ points (1999 HUN4)

Scores having at least 4 points in each problem and with only one 7 point:

$4 + 4 + 5 + 6 + 7 + 4 = 30$ points (1992 SWE1)

Problem Statistics

This part is about some statistics regarding the problems in the IMO. As usual, P1 denotes problem 1, etc.

Firstly, we provide the distribution of topics in the IMO problems.

Notes:

- Easy, Medium and Hard denote the numbers of times that a particular topic appears as an easy problem (P1 or P4), a medium problem (P2 or P5), and a hard problem (P3 or P6).

Since 2000:

Topic	Easy	Medium	Hard	Total
Algebra	9	16	9	34
Combinatorics	9	11	15	35
Geometry	20	10	14	44
Number Theory	10	11	10	31

Since 2014 (the recent 10 years):

Topic	Easy	Medium	Hard	Total
Algebra	3	7	2	12
Combinatorics	4	6	8	18
Geometry	8	2	7	17
Number Theory	5	5	3	13

Next, we see some statistics related to the difficulties of the problems.

The most number of contestants obtaining a particular score in a problem since 2000:

Score	P1	P2	P3	P4	P5	P6
0	207	522	608	218	451	562
1	96	213	145	145	255	126
2	111	129	80	106	174	57
3	105	138	31	74	124	39
4	72	79	40	128	50	19
5	54	53	9	69	35	9
6	121	129	9	68	49	7
7	474	303	73	433	250	56

The corresponding years for these to happen:

Score	P1	P2	P3	P4	P5	P6
0	2005	2021	2017	2021	2017	2021
1	2013	2000	2002	2005	2015	2020
2	2003	2020	2004	2018	2023	2005
3	2007	2017	2012	2012	2004	2016
4	2015	2017	2020	2008	2016	2022
5	2017	2008	2019	2009	2022	2000 2001
6	2004	2002	2005	2006	2022	2015
7	2023	2022	2023	2022	2019	2005

The least number of contestants obtaining a particular score in a problem since 2000:

Score	P1	P2	P3	P4	P5	P6
0	26	50	249	18	93	289
1	9	29	3	2	12	2
2	5	0	0	0	0	1
3	0	2	0	0	2	0
4	0	0	0	0	0	0
5	2	1	0	0	1	0
6	1	1	0	1	1	0
7	141	16	2	81	30	3

The corresponding years for these to happen (except when there are no contestants obtaining that particular score, since there are usually numerous of years for these to happen in those cases):

Score	P1	P2	P3	P4	P5	P6
0	2023	2000	2004	2006	2001	2004
1	2006	2020	2017	2010	2021	2009
2	2015 2020	-	-	-	-	2009 2020
3	-	2011 2021	-	-	2020	-
4	-	-	-	-	-	-
5	2004	2007 2021	-	-	2000 2003	-
6	2003	2003	-	2022	2008 2017	-
7	2005	2021	2007 2017	2000	2015	2009

Problems with the largest average scores since 2000:

2017 P1 (5.943 points)

2023 P1 (5.845 points)

2007 P4 (5.681 points)

2012 P1 (5.625 points)

2006 P1 (5.614 points)

Problems with the smallest average scores since 2000:

2017 P3 (0.042 points)

2007 P6 (0.152 points)

2009 P6 (0.168 points)

2006 P6 (0.187 points)

2016 P3 (0.251 points)

Apart from the nature and the difficulty of a problem, the length of the question statement is usually a factor to judge whether the problem is nice. Problems in the old days were usually shorter, while in recent years, the problems are usually written in an elaborated way so as to ensure that everyone understands the question.

Shortest problems in the IMO:

- (1963 Problem 5)

Prove that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$.

- (1962 Problem 4)

Solve the equation $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$.

- (1990 Problem 3)

Determine all integers $n > 1$ such that $\frac{2^n + 1}{n^2}$ is an integer.

Shortest problems in the IMO since 2000:

- (2006 Problem 4)

Determine all pairs (x, y) of integers such that $1 + 2^x + 2^{2x+1} = y^2$.

- (2022 Problem 5)

Find all triples (a, b, p) of positive integers with p prime and $a^p = b! + p$.

- (2007 Problem 5)

Let a and b be positive integers. Show that if $4ab - 1$ divides $(4a^2 - 1)^2$, then $a = b$.

Longest problem in the IMO: (2012 Problem 3)

The *liar's guessing game* is a game played between two players A and B . The rules of the game depend on two positive integers k and n which are known to both players.

At the start of the game A chooses integers x and N with $1 \leq x \leq N$. Player A keeps x secret, and truthfully tells N to player B . Player B now tries to obtain information about x by asking player A questions as follows: each question consists

of B specifying an arbitrary set S of positive integers (possibly one specified in some previous question), and asking A whether x belongs to S . Player B may ask as many such questions as he wishes. After each question, player A must immediately answer it with *yes* or *no*, but is allowed to lie as many times as she wants; the only restriction is that, among any $k + 1$ consecutive answers, at least one answer must be truthful.

After B has asked as many questions as he wants, he must specify a set X of at most n positive integers. If x belongs to X , then B wins; otherwise, he loses. Prove that:

1. If $n \geq 2^k$, then B can guarantee a win.
2. For all sufficiently large k , there exists an integer $n \geq 1.99^k$ such that B cannot guarantee a win.

In total, there are 181 words, 30 symbols and 4 short equations.

Very often, the problem statements appearing in the IMO are different from those in the shortlist, because the leaders would propose to modify the statements during the jury meetings. For example, 2007 P5 above is just a lemma used in N6 of the shortlist. The problem statement of 2012 C6, which was selected as 2012 P3, is much shorter. Let's see the comparison of problems in the shortlist since 2006, the first year when the IMO shortlist is available on the official website.

Shortest problems in the IMO shortlist since 2006:

- (2006 N5)

Find all integer solutions of the equation $\frac{x^7 - 1}{x - 1} = y^5 - 1$.

- (2022 N2)

Find all positive integers n such that $n! \mid \prod_{\substack{p < q \leq n, \\ p, q \text{ primes}}} (p + q)$.

- (2022 N8)

Prove that $5^n - 3^n$ is not divisible by $2^n + 65$ for any positive integer n .

Longest problems in the IMO shortlist since 2006:

- (2010 C6) 183 words, 25 symbols, 2 short equations

Given a positive integer k and other two integers $b > w > 1$. There are two strings of pearls, a string of b black pearls and a string of w white pearls. The *length* of a string is the number of pearls on it.

One cuts these strings in some steps by the following rules. In each step:

- (i) The strings are ordered by their lengths in a non-increasing order. If there are some strings of equal lengths, then the white ones precede the black ones. Then k first ones (if they consist of more than one pearl) are chosen; if there are less than k strings longer than 1, then one chooses all of them.
- (ii) Next, one cuts each chosen string into two parts differing in length by at most one.

(For instance, if there are strings of 5, 4, 4, 2 black pearls, strings of 8, 4, 3 white pearls and $k = 4$, then the strings of 8 white, 5 black, 4 white and 4 black pearls are cut into the parts (4, 4), (3, 2), (2, 2) and (2, 2), respectively.)

The process stops immediately after the step when a first isolated white pearl appears. Prove that at this stage, there will still exist a string of at least two black pearls.

- (2014 C6) 182 words, 24 symbols, 3 short equations

We are given an infinite deck of cards, each with a real number on it. For every real number x , there is exactly one card in the deck that has x written on it. Now two players draw disjoint sets A and B of 100 cards each from this deck. We would like to define a rule that declares one of them a winner. This rule should satisfy the following conditions:

1. The winner only depends on the relative order of the 200 cards: if the cards are laid down in increasing order face down and we are told which card belongs to which player, but not what numbers are written on them, we can still decide the winner.
2. If we write the elements of both sets in increasing order as $A = \{a_1, a_2, \dots, a_{100}\}$ and $B = \{b_1, b_2, \dots, b_{100}\}$, and $a_i > b_i$ for all i , then A beats B .
3. If three players draw three disjoint sets A, B, C from the deck, A beats B and B beats C , then A also beats C .

How many ways are there to define such a rule? Here, we consider two rules as different if there exist two sets A and B such that A beats B according to one rule, but B beats A according to the other.

- (2015 C1) 181 words, 13 symbols, 1 short equation

In Lineland there are $n \geq 1$ towns, arranged along a road running from left to right. Each town has a *left bulldozer* (put to the left of the town and facing left) and a *right bulldozer* (put to the right of the town and facing right). The sizes of the $2n$ bulldozers are distinct. Every time when a right and a left bulldozer confront each other, the larger bulldozer pushes the smaller one off the road. On the other hand, the bulldozers are quite unprotected at their rears; so, if a bulldozer reaches the rear-end of another one, the first one pushes the second one off the road, regardless of their sizes.

Let A and B be two towns, with B being to the right of A . We say that town A can *sweep* town B *away* if the right bulldozer of A can move over to B pushing

off all bulldozers it meets. Similarly, B can sweep A away if the left bulldozer of B can move to A pushing off all bulldozers of all towns on its way.

Prove that there is exactly one town which cannot be swept away by any other one.

4 Statistics of the Hong Kong Team

Here are various good and bad records of the Hong Kong team. Can you break any of these records?

First we go through some team results.

Highest total score of the whole team: 173 points (2022)

Lowest total score of the whole team: 68 points (1988)

Lowest total score of the whole team since 2000: 80 points (2000)

Highest score of the 2nd best team member since 2000: 32 points (2022)

Highest score of the 3rd best team member since 2000: 31 points (2016, 2022)

Highest score of the 4th best team member since 2000: 29 points (2022)

Highest score of the 5th best team member since 2000: 28 points (2022)

Highest score of the 6th best team member since 2000: 20 points (2014)

Highest rank of the whole team: rank 9 (2016)

Lowest rank of the whole team: rank 49 (2018)

Highest average individual rank: rank 72.83 (2016)

Highest rank of the 2nd best team member since 2000: rank 23 (2016)

Highest rank of the 3rd best team member since 2000: rank 23 (2016)

Highest rank of the 4th best team member since 2000: rank 63 (2021)

Highest rank of the 5th best team member since 2000: rank 93 (2007)

Highest rank of the 6th best team member since 2000: rank 145 (2007)

Most number of gold medals in one IMO: 3 (2016)

Most number of silver medals in one IMO: 5 (2007)

Most number of bronze medals in one IMO: 5 (1997, 2013)

Most number of HMs in one IMO: 4 (2018)

Highest score without medals or HM: 16 points (2019 HKG1)

Problems with perfect scores: 1995 P1, 2004 P4, 2006 P1, 2007 P2, 2007 P4, 2010 P4, 2011 P1, 2012 P1, 2014 P1, 2016 P1, 2016 P4, 2017 P1, 2020 P1, 2023 P1, 2023 P4

Highest total score in P1 since 2000: 42

Highest total score in P2 since 2000: 42 (2007)

Highest total score in P3 since 2000: 20 (2023)

Highest total score in P4 since 2000: 42

Highest total score in P5 since 2000: 36 (2022)

Highest total score in P6 since 2000: 16 (2005)

Lowest total score in P1 since 2000: 23 (2000, 2004)

Lowest total score in P2 since 2000: 3 (2019, 2021)

Lowest total score in P3 since 2000: 0

Lowest total score in P4 since 2000: 14 (2018)

Lowest total score in P5 since 2000: 8 (2006)

Lowest total score in P6 since 2000: 0

Most number of times of being the team leader: 12 (Leung Tat Wing)

Most number of times of being the deputy leader: 9 (Cesar Jose Alaban)

Most number of times of being the contestants: 4
(1998–2001 HKG1, 2000–2003 HKG6, 2005–2008 HKG6, 2014–2017 HKG6, 2020 HKG1–2023 HKG2)

Youngest team: 2020 (P6, S2, S3, S4, S5, S6 each, average: grade 9.33)

Problems in the shortlist proposed by HKG since 2006: 2010 G4 (2010 P2), 2014 N4, 2021 N7

The following are some individual results.

Best individual scores:

37 points (1989 HKG3, 1994 HKG5, 2005 HKG5)
36 points (1995 HKG1)
35 points (2009 HKG1, 2023 HKG2)
34 points (2022 HKG2)
32 points (2022 HKG5)
31 points (1989 HKG5, 1994 HKG2, 2016 HKG3, 2016 HKG5, 2016 HKG6, 2022 HKG3)

Lowest individual score: 1 point (2005 HKG6)

Highest individual ranks:

rank 14 (2009 HKG1)
rank 21 (1989 HKG3)
rank 23 (2005 HKG5, 2016 HKG3, 2016 HKG5, 2016 HKG6)
rank 25 (2011 HKG1, 2011 HKG3)
rank 27 (2006 HKG4)
rank 28 (2023 HKG2)
rank 29 (2002 HKG6)

Best total scores:

114 points (2020 HKG1–2023 HKG2)
94 points (2000–2003 HKG6)
91 points (2009–2011 HKG1)
89 points (2014–2017 HKG6)
82 points (2020 HKG6–2022 HKG5)
81 points (2007 HKG3–2009 HKG4)
79 points (1994 HKG2–1996 HKG1)
77 points (2019 HKG3–2021 HKG4)

Best total scores in algebra since 2000:

34 points (2009–2011 HKG1)
26 points (2007 HKG3–2009 HKG4)
25 points (2020 HKG1–2023 HKG2)
24 points (2000–2003 HKG6, 2011 HKG3–2012 HKG4)

Best total scores in combinatorics since 2000:

37 points (2014–2017 HKG6)

36 points (2020 HKG1–2023 HKG2)

33 points (2019 HKG3–2021 HKG4)

27 points (2000–2003 HKG6)

24 points (2015 HKG3–2017 HKG2)

Best total scores in geometry since 2000:

35 points (2005 HKG3–2007 HKG2)

34 points (2007 HKG3–2009 HKG4, 2008 HKG1–2010 HKG2)

32 points (2020 HKG1–2023 HKG2)

28 points (2005–2008 HKG6, 2009–2010 HKG3, 2009–2011 HKG1)

Best total scores in number theory since 2000:

27 points (2005 HKG5–2006 HKG4)

24 points (2014–2017 HKG6, 2020 HKG6–2022 HKG5)

23 points (2000–2003 HKG6)

Most number of gold medals: 3 (2009–2011 HKG1)

Most number of silver medals: 3 (2000, 2001, 2003 HKG6, 2005 HKG3–2007 HKG2, 2007 HKG3–2009 HKG4, 2019 HKG3–2021 HKG4, 2020 HKG6–2022 HKG5)

Most number of bronze medals: 3 (2020 HKG3–2022 HKG4)

Most number of HMs: 2 (1999 HKG6–2000 HKG5)

Most number of problems with 7 points in one IMO: 5 (1995 HKG1, 2005 HKG5, 2009 HKG1)

Most number of problems with 7 points: 12 (2020 HKG1–2023 HKG2)

Most number of problems with 6 points in one IMO: 2 (many examples)

Most number of problems with 5 points in one IMO: 3 (1994 HKG2)

Most number of problems with 4 points in one IMO: 3 (1993 HKG4)

Most number of problems with 3 points in one IMO: 3 (1990 HKG3)

Most number of problems with 2 points in one IMO: 2 (many examples)

Most number of problems with 1 point in one IMO: 5 (1988 HKG5)

Most number of problems with 0 points in one IMO: 5 (many examples)

Perfect scorers on one day:

1989 HKG3 (Day 2), 1994 HKG3 (Day 1), 1994 HKG5 (Day 1), 1995 HKG1 (Day 1),
2001 HKG3 (Day 1), 2005 HKG3 (Day 2), 2005 HKG5 (Day 1), 2009 HKG1 (Day 1),
2023 HKG2 (Day 1)

Team members solving P3 or P6 since 2000:

2001 HKG3 (P3), 2004 HKG5 (P6), 2005 HKG1 (P3), 2005 HKG3 (P6),
2005 HKG5 (P3), 2009 HKG1 (P3), 2011 HKG1 (P3), 2011 HKG2 (P6),
2011 HKG3 (P3), 2015 HKG6 (P6), 2020 HKG4 (P3), 2021 HKG2 (P3),
2022 HKG2 (P6), 2023 HKG2 (P3)

Team members solving a hard problem with less than 50 solvers:

2011 HKG2 (P6 geometry, 6 solvers)
2015 HKG6 (P6 combinatorics, 11 solvers)
2021 HKG2 (P3 geometry, 15 solvers)
2001 HKG3 (P3 combinatorics, 20 solvers)
2022 HKG2 (P6 combinatorics, 22 solvers)
2010 HKG4 (P5 combinatorics, 37 solvers)
2020 HKG4 (P3 combinatorics, 42 solvers)
2004 HKG5 (P6 number theory, 48 solvers)
2006 HKG4 (P5 number theory, 48 solvers)

Youngest gold medalist: 2021 HKG2 (S1)

Youngest silver medalist: 2020 HKG2 (P6)

Youngest bronze medalist: 2020 HKG3 (S2), 2023 HKG1 (G8)

Youngest HM awardee: 2012 HKG3 (S3)

Best improvement in the total score compared to the previous IMO:

1993 HKG5 (12 points) $\rightarrow$ 1994 HKG5 (37 points)
2021 HKG6 (4 points) $\rightarrow$ 2022 HKG6 (29 points)

Best improvement in the rank compared to the previous IMO:

2021 HKG6 (rank 439) $\rightarrow$ 2022 HKG6 (rank 112)

Scores forming a strictly decreasing sequence: $7 + 6 + 5 + 4 + 2 + 0 = 24$ (2002 HKG3)

Scores forming a palindrome: $7 + 3 + 0 + 0 + 3 + 7 = 20$ (2015 HKG6)

Curse on the gold medalists of the Hong Kong team: Every time when a team member obtains his first gold medal, his result in the next IMO will become worse.

2002 HKG6 (29 points, rank 29, gold) → 2003 HKG6 (19 points, rank 98, silver)
2005 HKG5 (37 points, rank 23, gold) → 2006 HKG4 (28 points, rank 27, gold)
2009 HKG1 (35 points, rank 14, gold) → 2010 HKG1 (27 points, rank 42, gold)
2011 HKG3 (29 points, rank 25, gold) → 2012 HKG4 (25 points, rank 64, silver)
2016 HKG3 (31 points, rank 23, gold) → 2017 HKG2 (25 points, rank 36, gold)
2016 HKG6 (31 points, rank 23, gold) → 2017 HKG6 (17 points, rank 188, bronze)
2016 HKG5 (31 points, rank 23, gold) → decide not to join the IMO team anymore
2021 HKG2 (28 points, rank 32, gold) → 2022 UNK1 (31 points, rank 68, silver)

Exception:

2022 HKG2 (34 points, rank 38, gold) → 2023 HKG2 (35 points, rank 28, gold)

Maybe we are beginning a new trend of the pattern? Or perhaps one day we can replace 'his' to 'his/her'?

Easter Egg

There are two more secrets related to the gold medals of the Hong Kong team given below. Are you clever enough to read the secrets? If yes, please keep these confidential like the IMO shortlist!

012020102122: 122012002 202110120001121 220120002 202120020122122002102
020102 202012020122002 222110100110200120